AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 18 · STUDENT LESSON

Retrieval Augmented Generation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Retrieve. Inspect. Draft. Cite. Check each claim.

Ground an answer in supplied passages without inventing missing details.

Keep one habit: Evidence map → ice note → four-passage paper RAG → missing duration → independent conditional room case.

MAIN LESSON · SECTION 1

Can every answer claim point to its evidence?

> Connect retrieval and generation while preserving missing information.

Your main learning goals

I can explain the **mechanism** and terminology of building a **grounded answer**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall and preview the evidence route

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Recall Source Quality: identify the question, relevant time and authority before selecting a passage.
- Recall Model Memory vs Retrieval: an inspected passage is different from generated wording. A missing relevant source leaves a factual gap.
- Preview **chunk**ing: keep a rule with its conditions and exceptions. Preview ranking: a similarity score is not a source-quality guarantee. These two strands are explained in this week's extension.

Your end goal

By the end, I can explain retrieval augmented generation using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Model Memory vs Retrieval. Give an example and a mistake that the idea helps you avoid.

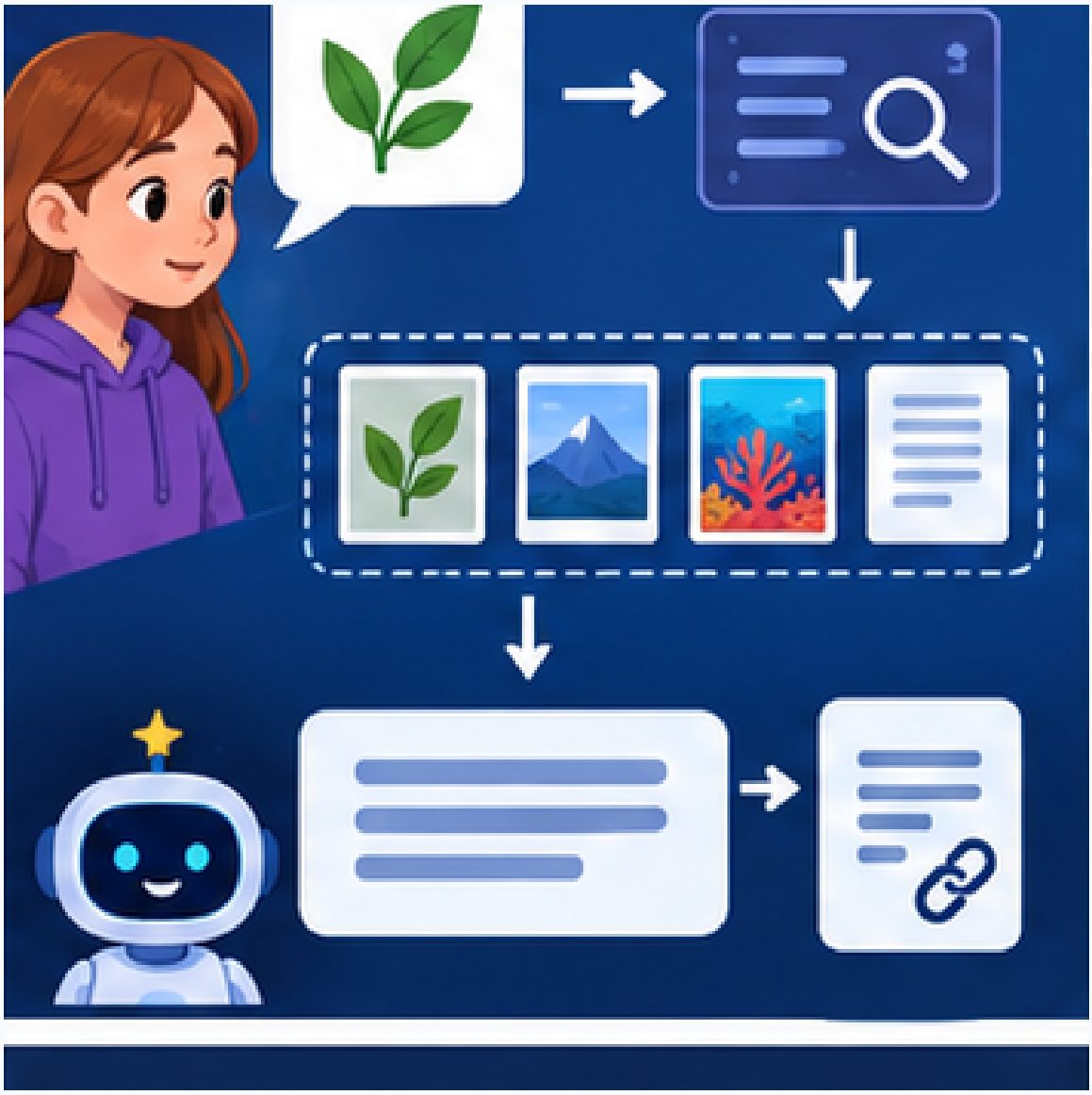
Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 18: Retrieval-Augmented Generation

RAG retrieves passages, then generates an answer grounded in them.



CREATOR MISSION

Trace **query** → retrieve → select → answer → cite.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Readiness for Chunking and document structure

My readiness explanation

Readiness for Lexical and vector retrieval

My readiness explanation

This week’s end goal

MAIN LESSON · SECTION 2

Retrieve evidence, then check what the answer adds

> The answer must stay inside the selected sources’ support.

Understand RAG and grounding

Retrieval augmented generation, or RAG, first retrieves relevant records and then uses them as context for a generated answer. Grounding means connecting each factual claim to evidence that actually supports it. A retrieved passage can be irrelevant, stale, incomplete or misused. Citations make checking possible; they do not perform the check for us. This lesson traces supplied paper records, not a live model.

A **grounded answer** connects each factual claim to **evidence** actually retrieved for the task. Retrieval success does not automatically make generation faithful.

▼ Extension · Chunking and ranking preview

Chunking divides documents into retrievable units. A useful **chunk** retains enough structure to interpret a passage, including headers and exceptions.

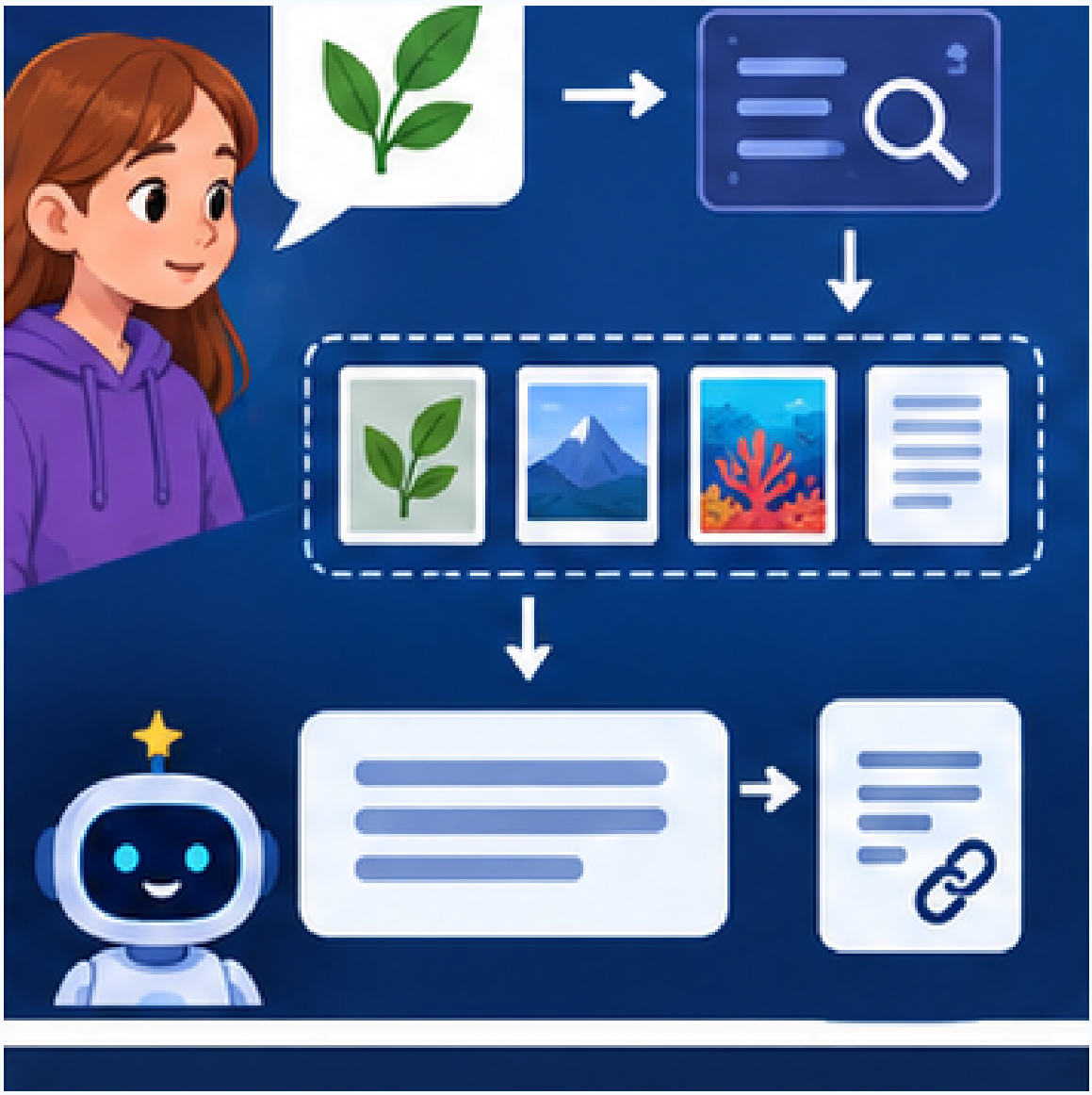
Lexical retrieval matches terms; **vector retrieval** compares numerical representations. Each can fail when the representation misses the relevant distinction.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use retrieval-augmented generation to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Chunking and document structure

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

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Apply the connected idea: Lexical and vector retrieval

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Build the claim-evidence map before final wording

> A source name without support is insufficient.

Trace the main evidence route

Stage	Evidence to preserve
Question	Entity, requested fact and relevant time
Retrieve	Eligible passage IDs, versions and scope
Inspect	Direct support, linked exceptions and gaps
Draft	Only supported factual claims; explicitly state missing answers
Cite and verify	Map each claim to its passage and check qualifiers again

Build a claim-**evidence table** before final wording. Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Extension · Chunk identity and linked exceptions

Give each **chunk** its document/version ID, section and **offset**s. Preserve **qualifier**s or link adjacent material. **Overlap** can repeat information; repeated **chunk**s are not independent sources.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ Extension · Exact toy lexical and vector rules

For a toy lexical score, count distinct **query** terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare **relevance** using independent judgments, not score magnitude alone.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

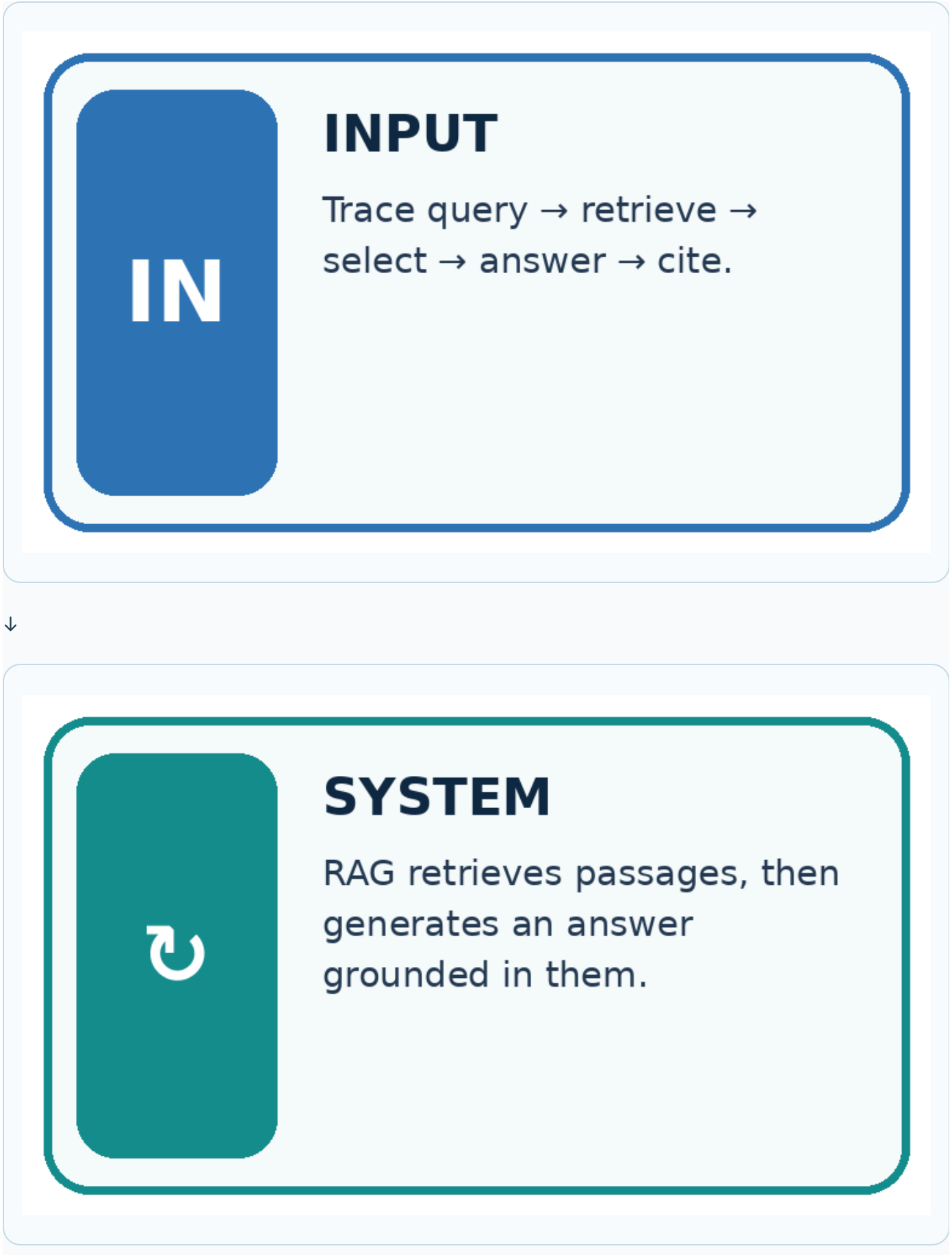
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ Earlier wording, alternatives and picture transcripts

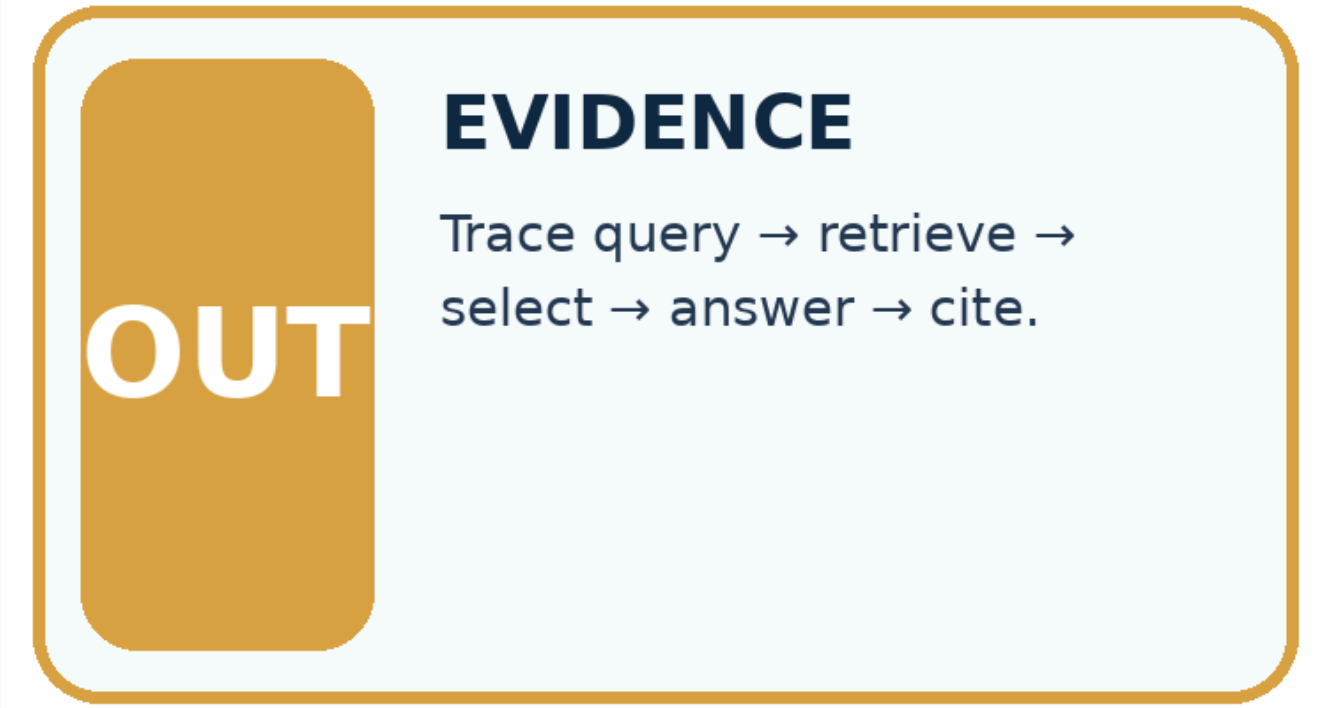
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Trace query → retrieve → select → answer → cite.
↓
↻
SYSTEM
RAG retrieves passages, then generates an answer grounded in them.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Building a grounded answer — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch.	Build a claim-evidence table before final wording. Preserve source versions and qualifiers, omit unsupported claims or explicitly state the gap, then validate the answer against the table.	Approval and capacity are supported by P and Q; lunch lacks evidence. A grounded answer reports the first two claims and does not invent lunch.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Apply the connected idea: Chunking and document structure

Chunking and document structure — evidence trace

INPUT AND SCOPE

A notice says capacity 20, followed by during maintenance capacity 8. Separate chunks contain each sentence but retrieval returns only the first.

CHECK OR CALCULATION

Give each chunk its document/version ID, section and offsets. Preserve qualifiers or link adjacent material. Overlap can repeat information; repeated chunks are not independent sources.

CONCLUSION AND LIMIT

The first chunk alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Apply the connected idea: Lexical and vector retrieval

Lexical and vector retrieval — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Query terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector query [1, 0], D1=[.4, .9], D2=[.8, .1].	For a toy lexical score, count distinct query terms present after lowercase exact-word matching. For a toy vector score use a stated dot product. Compare relevance using independent judgments, not score magnitude alone.	Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task relevance favours the sensor manual; vector ranking alone does not prove relevance.

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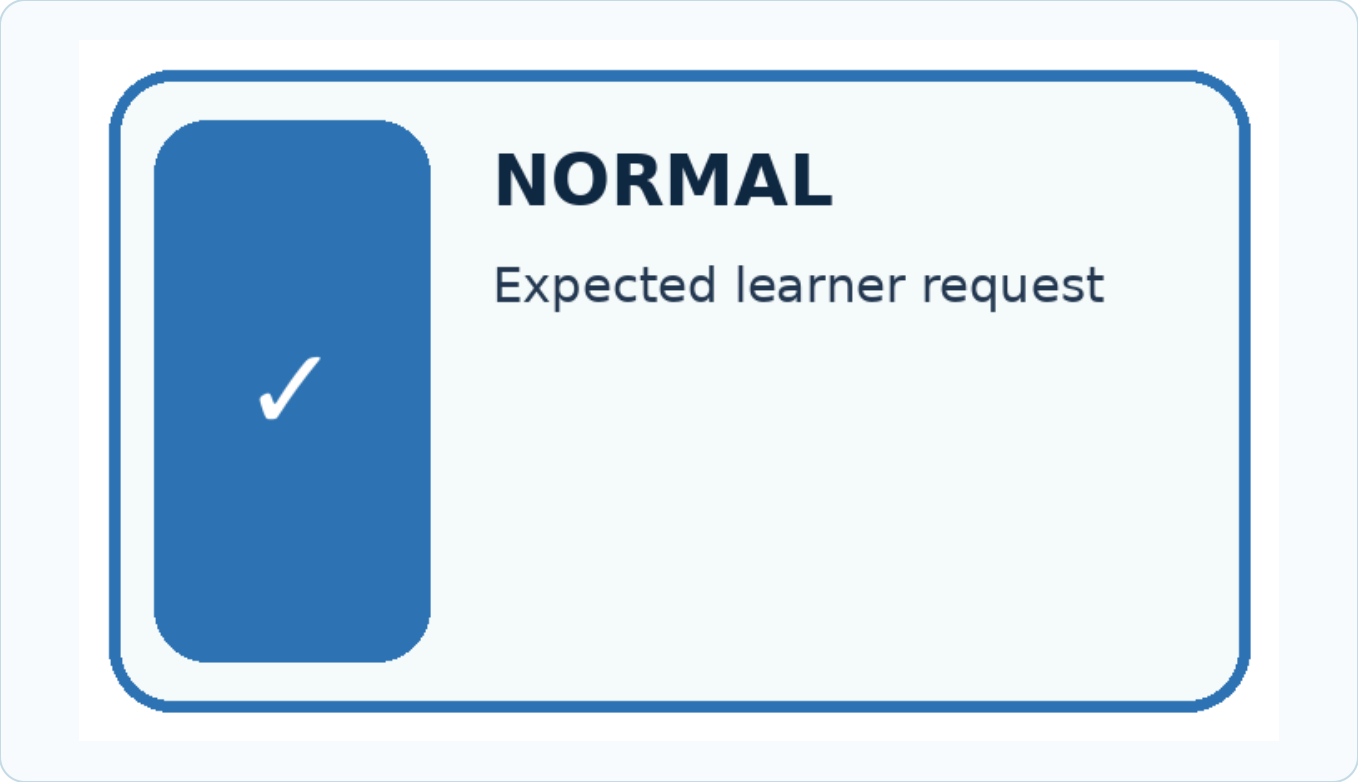
Connect the picture and the explanation

MAIN LESSON · SECTION 4

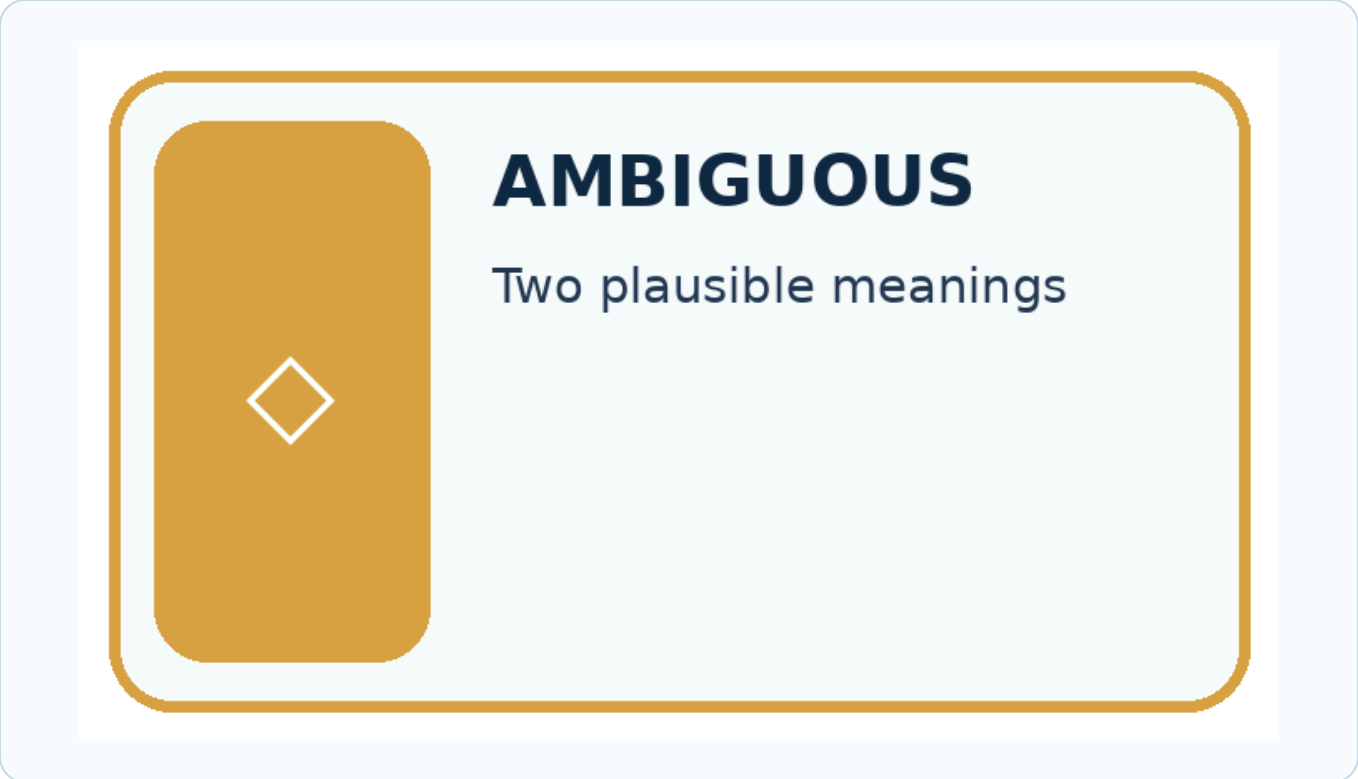
Compare grounded answers with unsupported additions

> Conditions and missing details must survive generation.

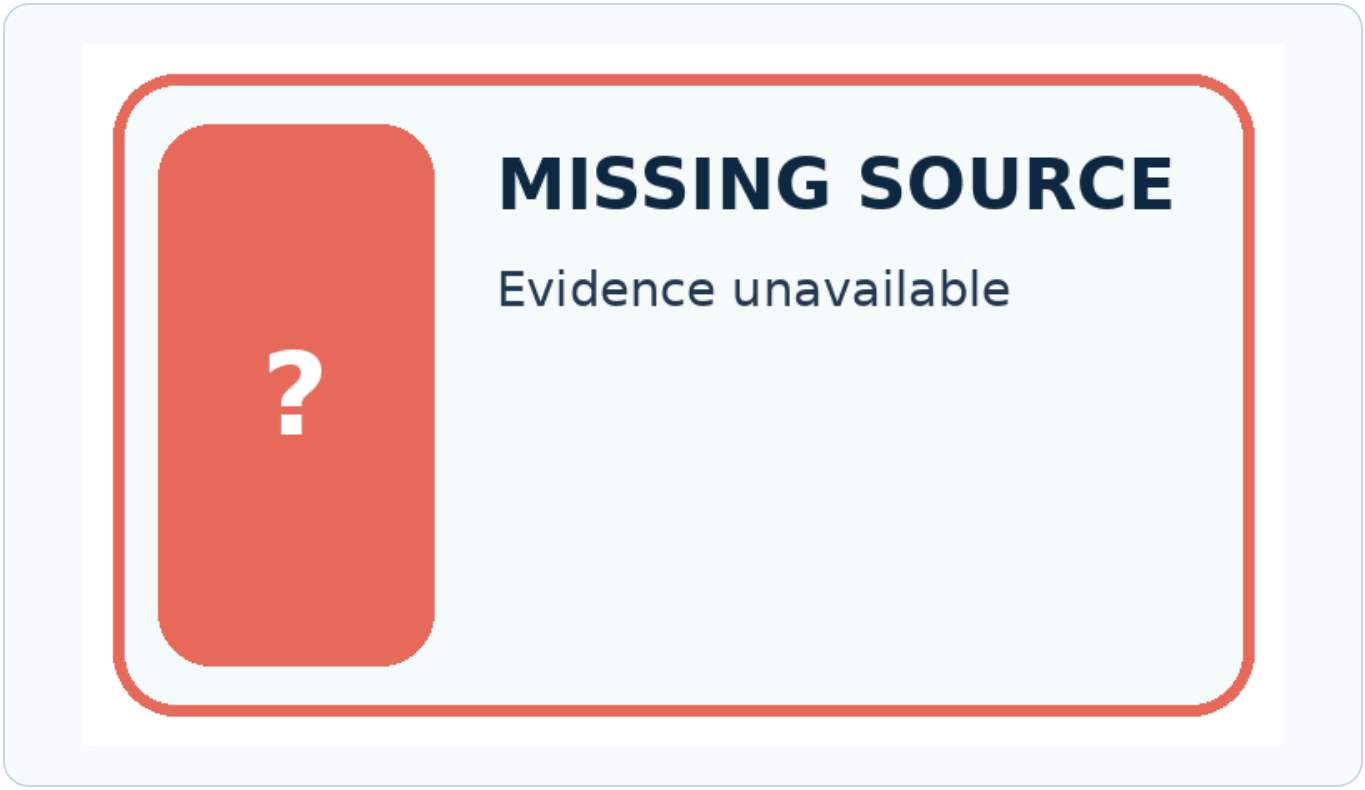
Four pictures with a purpose



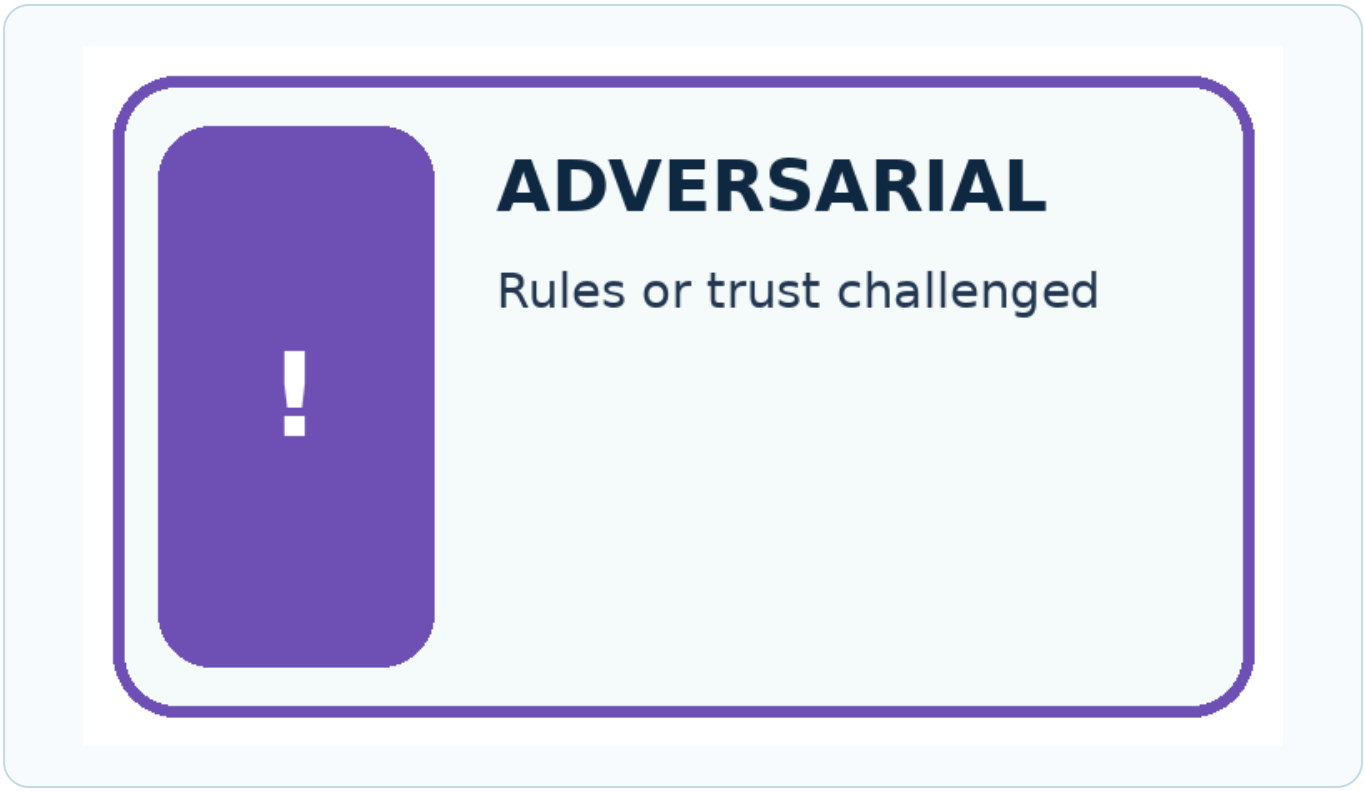
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Predict the supplied changed case

- Change: The question asks exactly how long this ice takes to melt.

▼ Check after predicting

Prediction: Say that S4 does not provide a time.

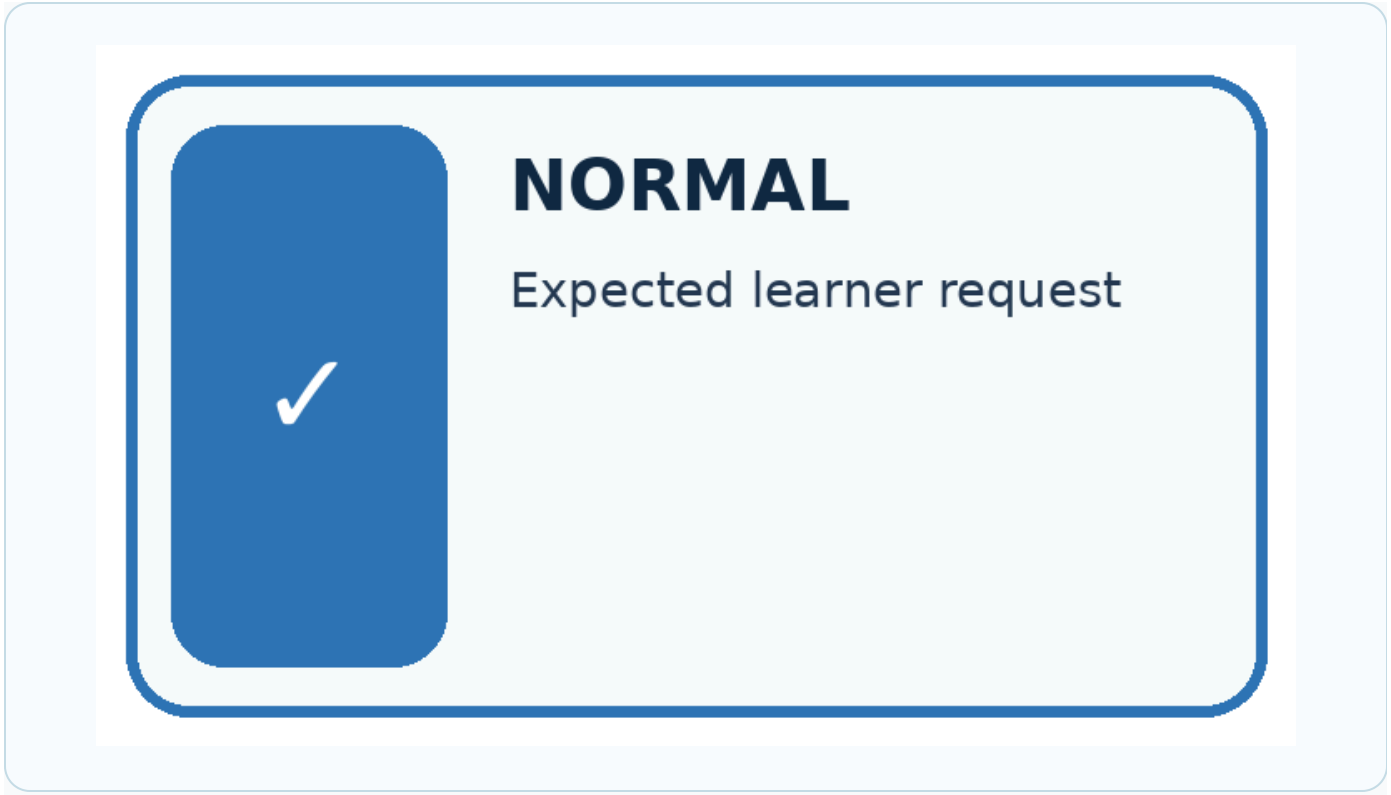
Reason: A general change of state does not determine a duration for an unspecified sample.

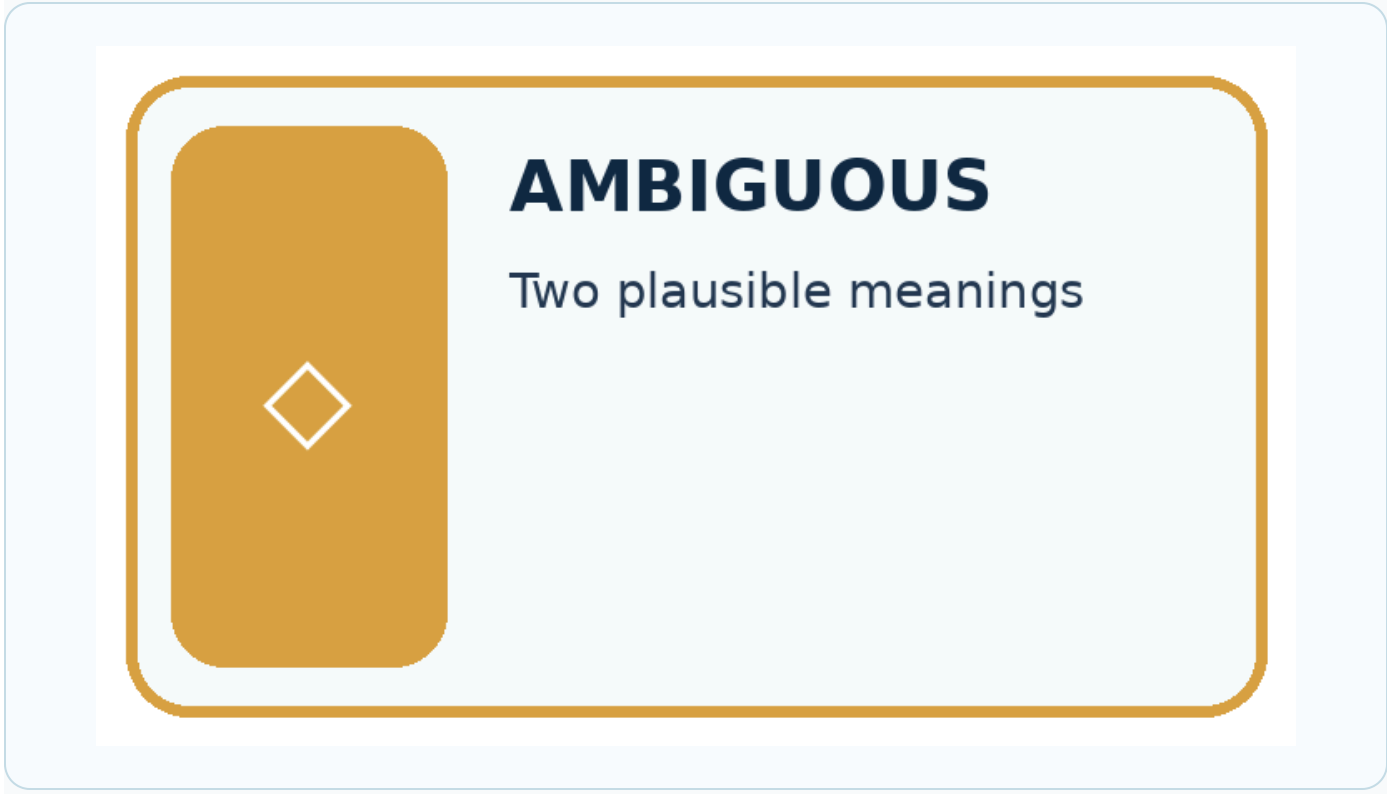
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the ice explanation back to S4

> A change of state does not determine a sample’s melting duration.

Read the complete source note

Turn a short science source into an answer with a visible claim-to-source map.

Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Map the two supported claims

AI ACADEMY / AI-3/18 · 02 · TRACE THE CASE

Follow the deciding steps

01 Read source

S4 supports the material identity and the solid-to-liquid change.

02 Draft two claims

Draft: ice is solid water; warming can turn it into liquid water.

03 Check each claim

Both claims match S4. A claim that melting takes exactly two minutes would not match.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read source	S4 supports the material identity and the solid-to-liquid change.
2. Draft two claims	Draft: ice is solid water; warming can turn it into liquid water.
3. Check each claim	Both claims match S4. A claim that melting takes exactly two minutes would not match.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

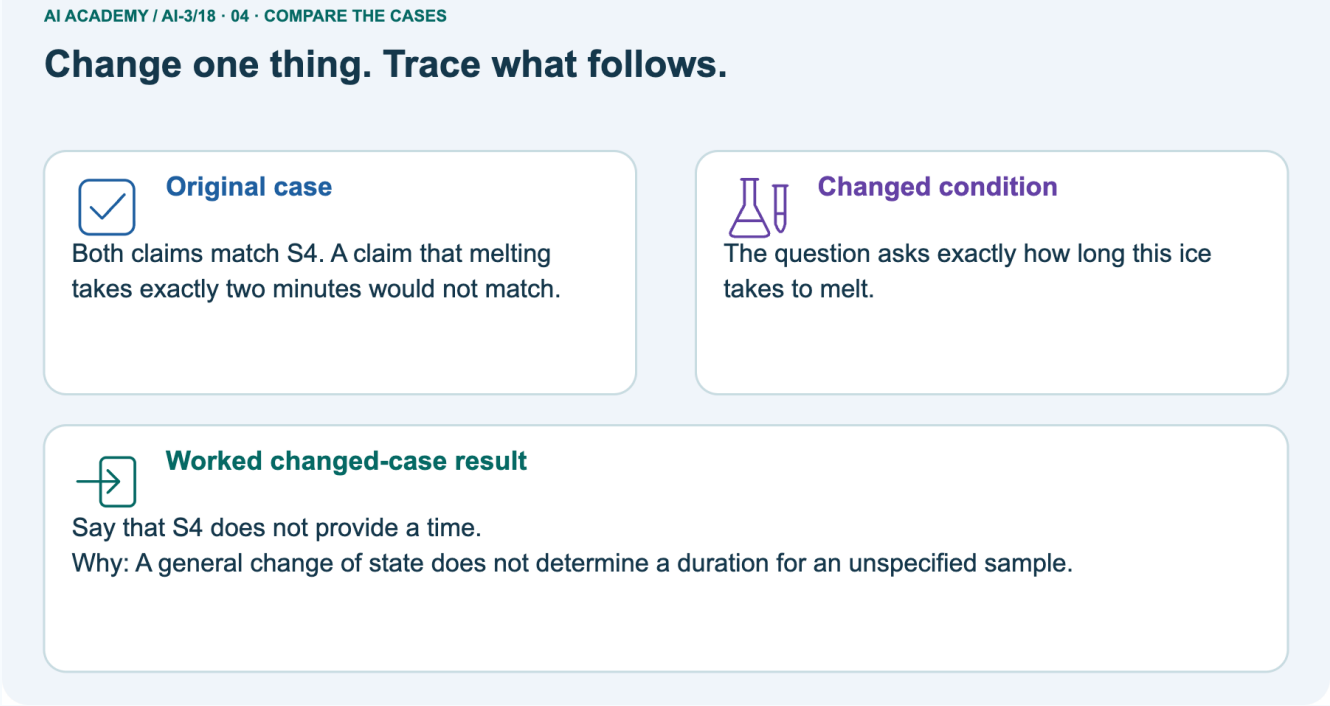
Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict the exact-duration question

- Change one thing: The question asks exactly how long this ice takes to melt. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State source scope and the missing detail

- This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

- Read the labelled picture: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate practice · Room C approval and capacity

Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch.

Approval and capacity are supported by P and Q; lunch lacks **evidence**. A **grounded answer** reports the first two claims and does not invent lunch.

▼ Extension · Preserve a maintenance exception

A notice says capacity 20, followed by during maintenance capacity 8. Separate **chunk**s contain each sentence but retrieval returns only the first.

The first **chunk** alone lacks the applicable exception. Retrieve the linked condition and determine maintenance status before asserting capacity 20.

▼ Extension · Compare exact toy ranking scores

Query terms red sensor. D1 text red sensor manual; D2 text red paint. Their lexical scores are 2 and 1. Vector **query** [1, 0], D1=[.4, .9], D2=[.8, .1].

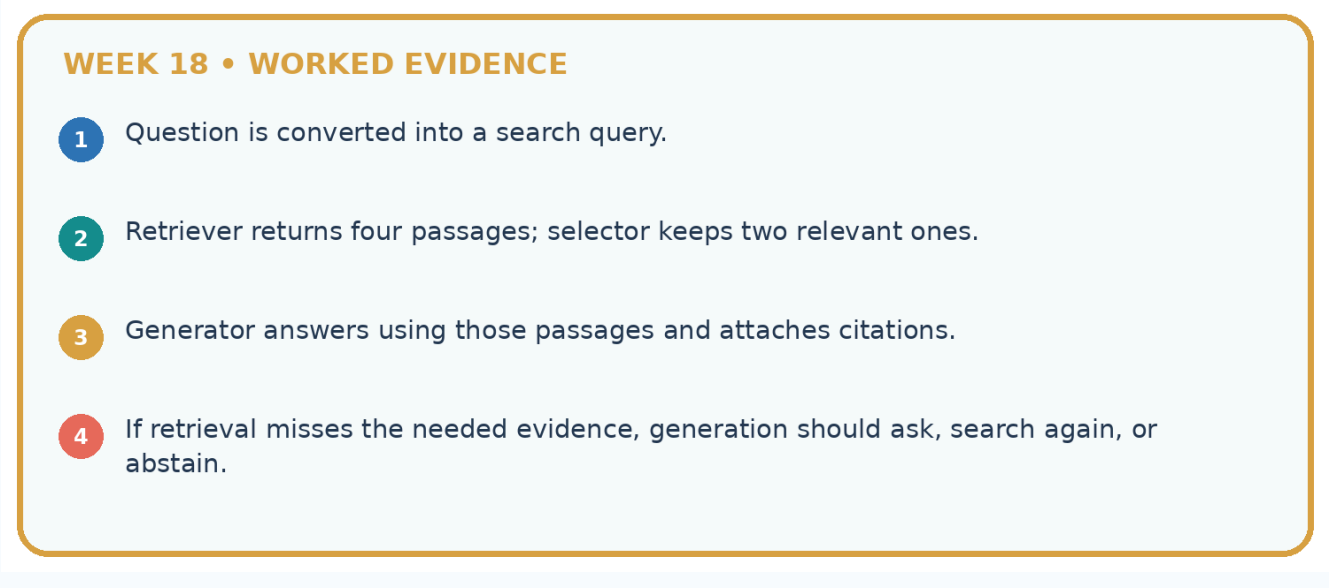
Lexical ranks D1 first; dot product ranks D2 first (.8 versus .4). The supplied task **relevance** favours the sensor manual; vector ranking alone does not prove **relevance**.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Question is converted into a search **query**.

2. **Retriever** returns four passages; selector keeps two relevant ones.

3. Generator answers using those passages and attaches citations.

4. If retrieval misses the needed **evidence**, generation should ask, search again, or abstain.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Chunking and document structure

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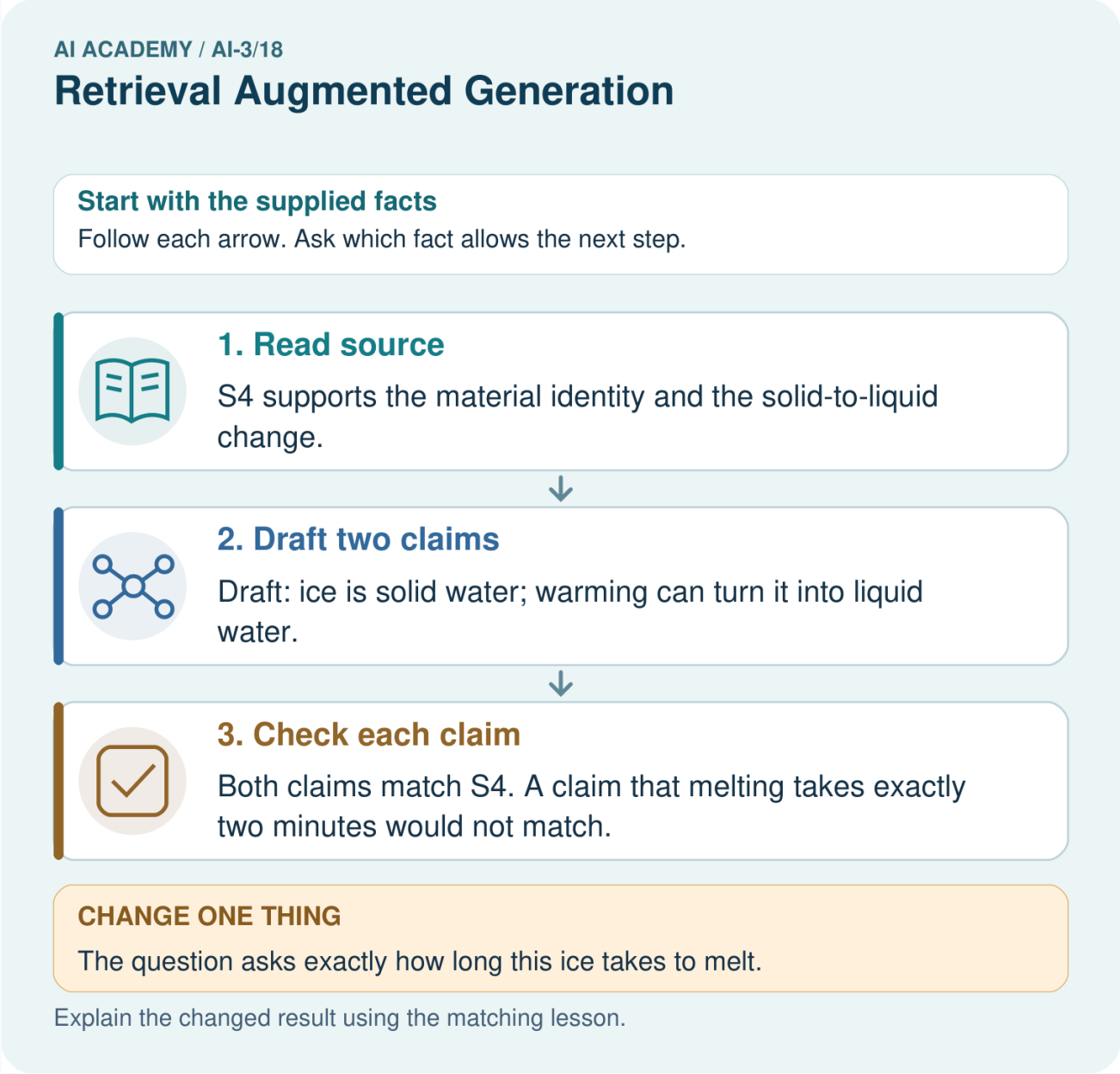
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Retrieval Augmented Generation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the unsupported detail after retrieval

> Good retrieval can still be followed by an ungrounded answer.

Repair the duration shortcut

Claim to repair: S4 says warming can melt ice, so the sample must melt in exactly two minutes. S4 gives no duration, sample size or warming conditions. Keep its supported material and state-change claims, but say the exact time is not supplied. Adding a citation to S4 does not make the extra duration supported. Ask for relevant evidence instead of guessing.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: RAG guarantees correctness because it has documents. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Apply the connected idea: Chunking and document structure

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

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Practise a grounded answer with less help

> Use room C for supported W2; keep the fresh room H task independent.

The grounding boundary

Grounding requires support for the exact claim, including conditions; attaching a source name is insufficient.

Complete the room C claim map

Use workbook W2 for the connected example “Building a **grounded answer**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for Building a **grounded answer**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured missing duration

- Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Keep the independent case separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Extension · Chunking tasks

Smaller **chunk**s can improve selectivity while losing necessary **qualifier**s; **chunk** size alone is not retrieval quality.

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

▼ Extension · Ranking tasks

Similarity and **relevance** are different quantities; exact terminology, metadata and permissions may be necessary.

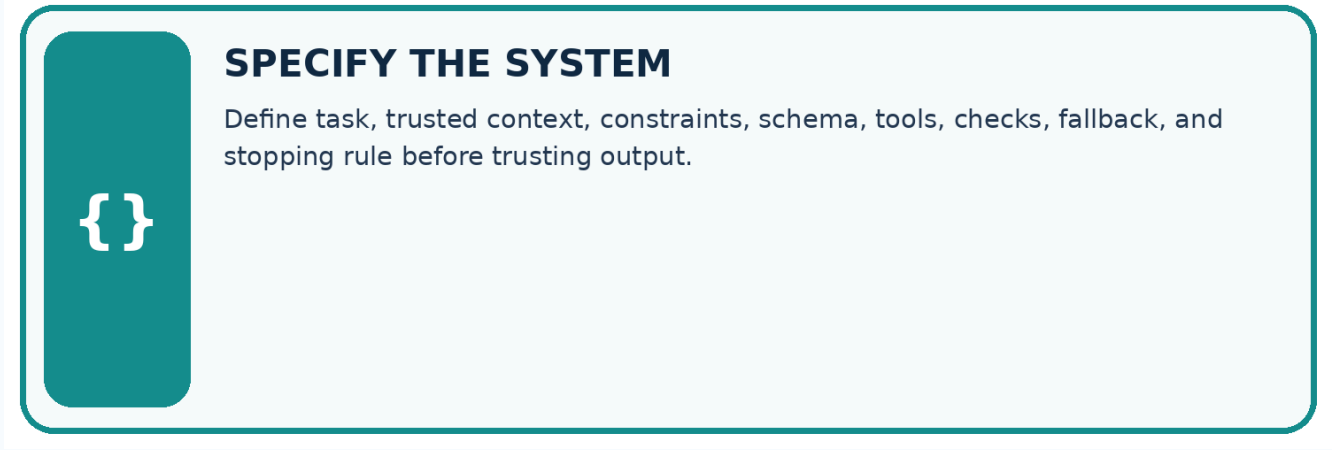
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Apply the connected idea: Chunking and document structure

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Practise with less help: Trace the worked case

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Practise the connected topic: Chunking and document structure

Practise the connected topic: Lexical and vector retrieval

Explain the picture

MAIN LESSON · SECTION 9

Run paper RAG with four complete passages

> Check relevance, then check each generated claim.

All supplied passage cards

ID / complete fictional passage	Role in the question
S4-v1 · Approved science note: ice is solid water; warming can melt ice into liquid water.	Supports material and change of state; supplies no duration.
R2-v1 · Classroom tidying note: return used cups to the blue tray.	Unrelated to ice identity, state change and duration.
G3-v1 · Fictional story: a wizard’s ice castle vanished in two minutes.	Explicit fiction; not a timing record for the asked sample.
M1-v1 · Measurement template: record sample size, start conditions, elapsed time and observation. No completed measurements.	Can guide a future test; supplies no observed result.

Predict, select, draft, cite and audit

1. Paper RAG task: answer What is ice, and what change can warming cause? Predict relevant passages, then inspect all four complete cards. Select S4-v1, map each of the two claims to its wording, and write a bounded answer with its ID. State why R2 is unrelated, G3 is fictional and M1 is an unfilled template. Missing-answer variation: ask exactly how long an unspecified ice sample takes to melt.

   No supplied card answers this; say the time is unavailable and identify needed measurement conditions. Do not count the fictional story as an experiment or invent a run. Keep prediction, paper trace and proposed future test separate. Retain room H for an independent attempt.

Keep the evidence trail

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

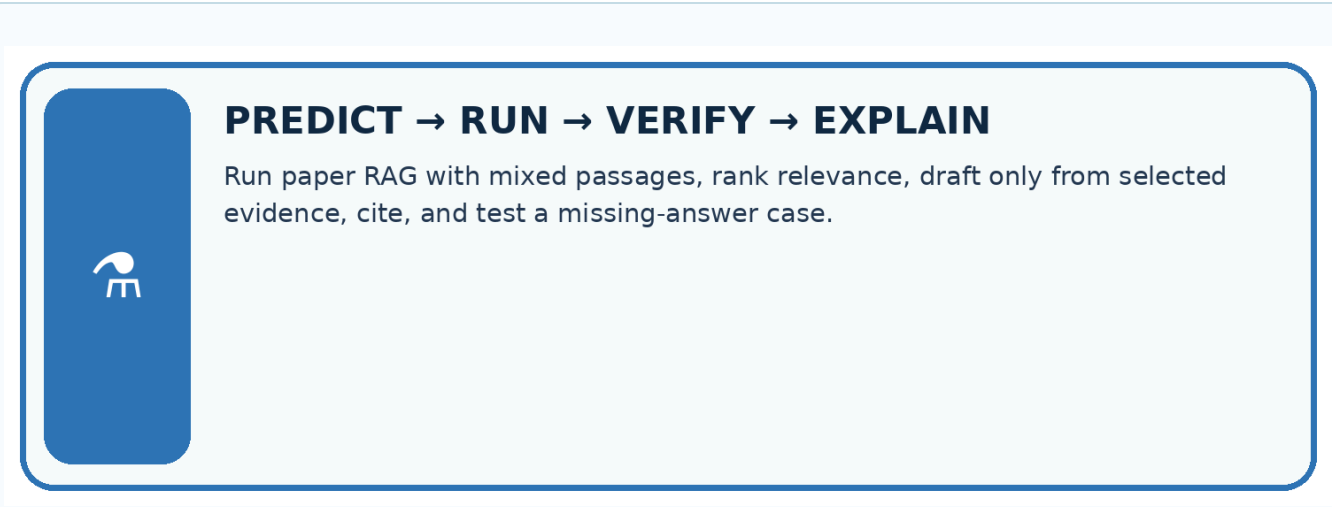
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the four complete S4/R2/G3/M1 passage cards for paper RAG. Rank by the exact question, draft only supported claims, cite the source version and test the missing-duration question. Record paper traces without claiming a live retrieval or model run.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: ground the fresh room H response

> Keep the first attempt separate from the room C practice.

Independent sources and scope

Source A approves room H; B lists H capacity 7, except maintenance limit 4. Maintenance status is unknown. Write a grounded response.

Use only the stated approval and conditional capacity sources. No maintenance-status record is provided. State what is supported and what remains unresolved without adding a new room inspection or source.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply a new case independently

Week 18 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A retrieved passage says the club meets Tuesday; it says nothing about cost. The assistant answers Tuesday, free entry. Which part is grounded and what should replace the unsupported part?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Implement question → retrieve → inspect → answer → cite → verify as the core architecture.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Sources: A='Library opens 9'; B='Cafe opens 8'; C='Library closes 5'. A question asks when the library opens. The **retriever** returns only B and C. What should the generator do?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project’s claim-to-source trail

> Inspect the passage before trusting the generated wording.

This week’s contribution

For the Verified Study Buddy, save the query, eligible retrieved passages, source versions, claim-evidence map, final answer and unresolved fields. Preserve linked exceptions. Check that a citation supports its exact claim rather than merely sharing a topic. If retrieved context misses the answer, ask for the relevant source or state the gap. A complete-looking response is not evidence of successful grounding.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK’S CHECKPOINT Implement question → retrieve → inspect → answer → cite → verify as the core architecture.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

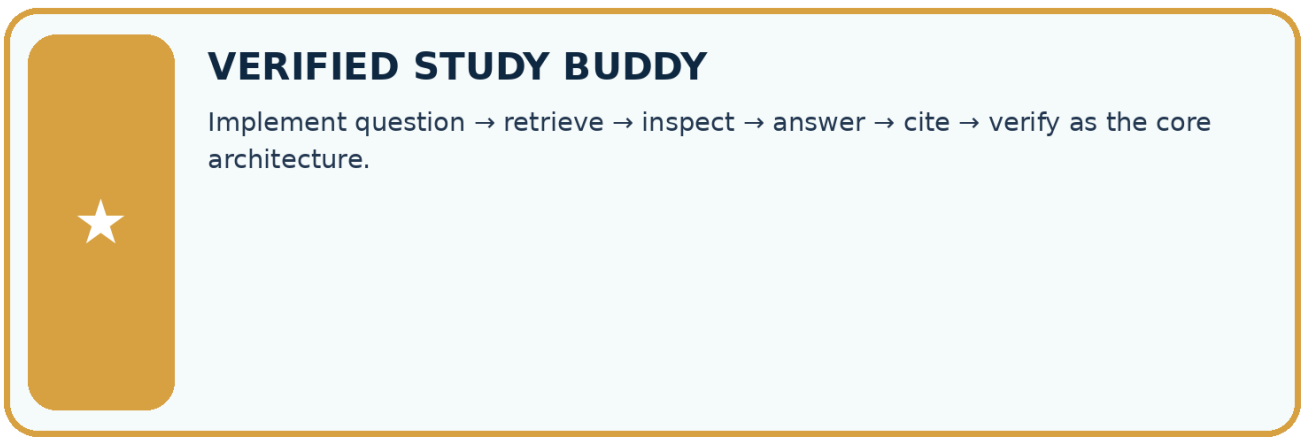
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: show the passage behind each factual claim

> A retrieved source can leave the requested detail unanswered.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of building a **grounded answer**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
query	A question or search request used to obtain information.	
retriever	The component that searches for and selects source material relevant to a query.	
relevance	How directly information helps answer the stated question or perform the task.	
grounded answer	An answer whose factual claims are supported by identified evidence relevant to the question.	

Main explanation terms

Word	Meaning
Grounded claim	A claim supported by the available task evidence

Word	Meaning
Qualifier	A condition restricting a statement
Evidence table	A mapping from each claim to its supporting record

Extension terms

Term	Meaning
Chunk	A retrievable unit from a document
Overlap	Content repeated in adjacent chunks
Offset	A location linking a chunk to the original source

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

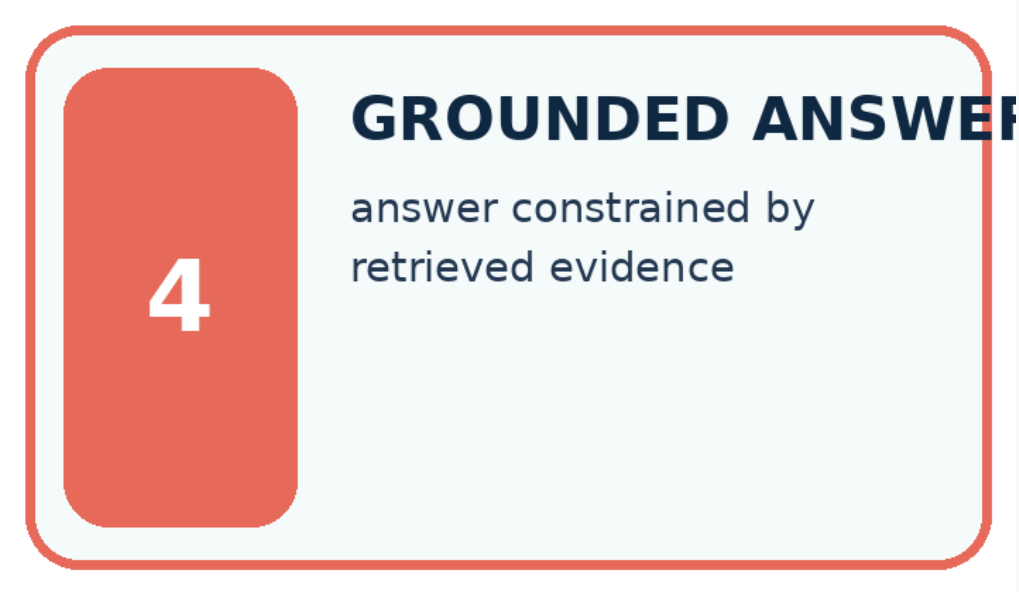
RETRIEVER

finds candidate passages

3

RELEVANCE

fit between passage and question



Words used in the closer explanation

Terms for Chunking and document structure

Terms for Lexical and vector retrieval

Term	Meaning
Lexical match	Matching query terms to stored text
Vector retrieval	Ranking numerical representations using a declared similarity
Relevance judgment	An assessment of usefulness for the particular query

▼ Extra challenge · Choose a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

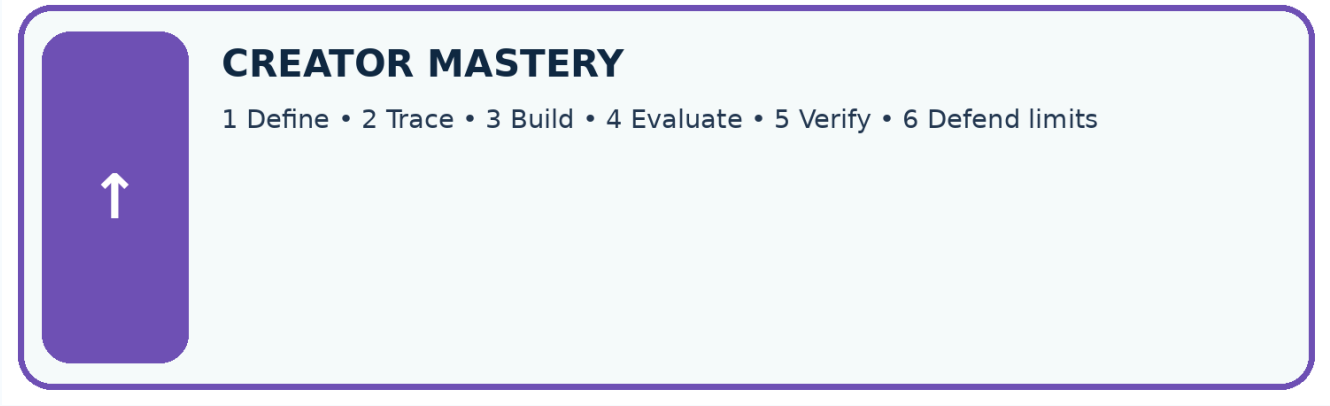
> **Write or draw your reasoning and evidence.**

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

REFERENCE · SECTION 12

Reference · Check your explanation

> Use after practice; Ground the new room H response independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check

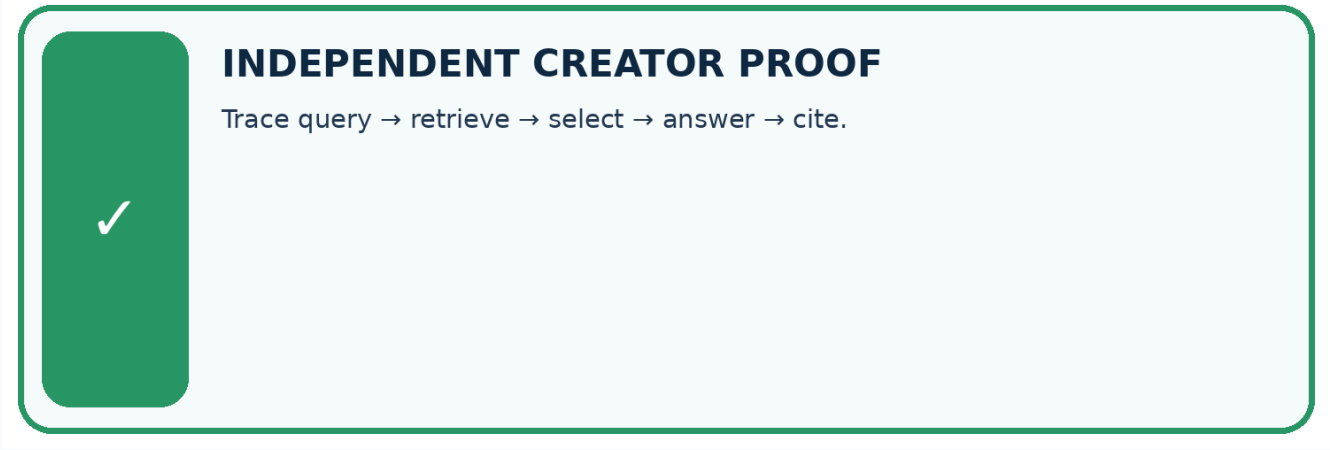
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week's explanation

Use after practice. Ground the new room H response independently is the selected independent task.



Require claim-level support, preserved conditions and an explicit unknown maintenance status.

▼ Extra practice · The earlier complete case

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

Note A says library opens at 9. Note B says returns go in the blue box.

Answer opening time with **evidence**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

Evidence or stage	Value or result
Question	When does the library open?
Retrieve	Note A says opens at 9
Compose	Library opens at 9
Verify	Note A supports opening time

▼ Earlier wording, alternatives and picture transcripts

Week 18 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Use the passage that answers the question

Note A says library opens at 9. Note B says returns go in the blue box.

Answer opening time with **evidence**.

Answer 9 with note A. Retrieval supplies **evidence** and generation phrases it. B does not support opening time.

▼ **Reference · Earlier checklist**

REFERENCE · SECTION 15

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Explain the mechanism and terminology of building a grounded answer	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Turn a short science source into an answer with a visible claim-to-source map.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- Turn a short science source into an answer with a visible claim-to-source map.
- Which labelled step matches this detail? Both claims match S4. A claim that melting takes exactly two minutes would not match.
- 1. Check each claim
- 2. Draft two claims
- 3. Read source

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Check each claim Both claims match S4. A claim that melting takes exactly two minutes would not match. Read the picture in order. Both claims match S4.

  A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after trying**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The question asks exactly how long this ice takes to melt. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. Say that S4 does not provide a time.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Say that S4 does not provide a time. A general change of state does not determine a duration for an unspecified sample.

> **My selected conclusion, deciding fact and explanation**

▼ **Check changed case**

Repair a missing building block

Repair and keep evidence

Review Model Memory vs Retrieval (ai-3/17). Goal: Explain the **mechanism** and terminology of retrieval and indexing. Short practice: Why can the assistant not certify 15: 00 when the notice is missing?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Source Quality (ai-3/15). Goal: Explain the **mechanism** and terminology of conflicting and time-sensitive sources. Short practice: Which source fields must be checked before answering day 9?.

Return to this week and explain what you changed.

Review Speech and Audio Systems (ai-3/12). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

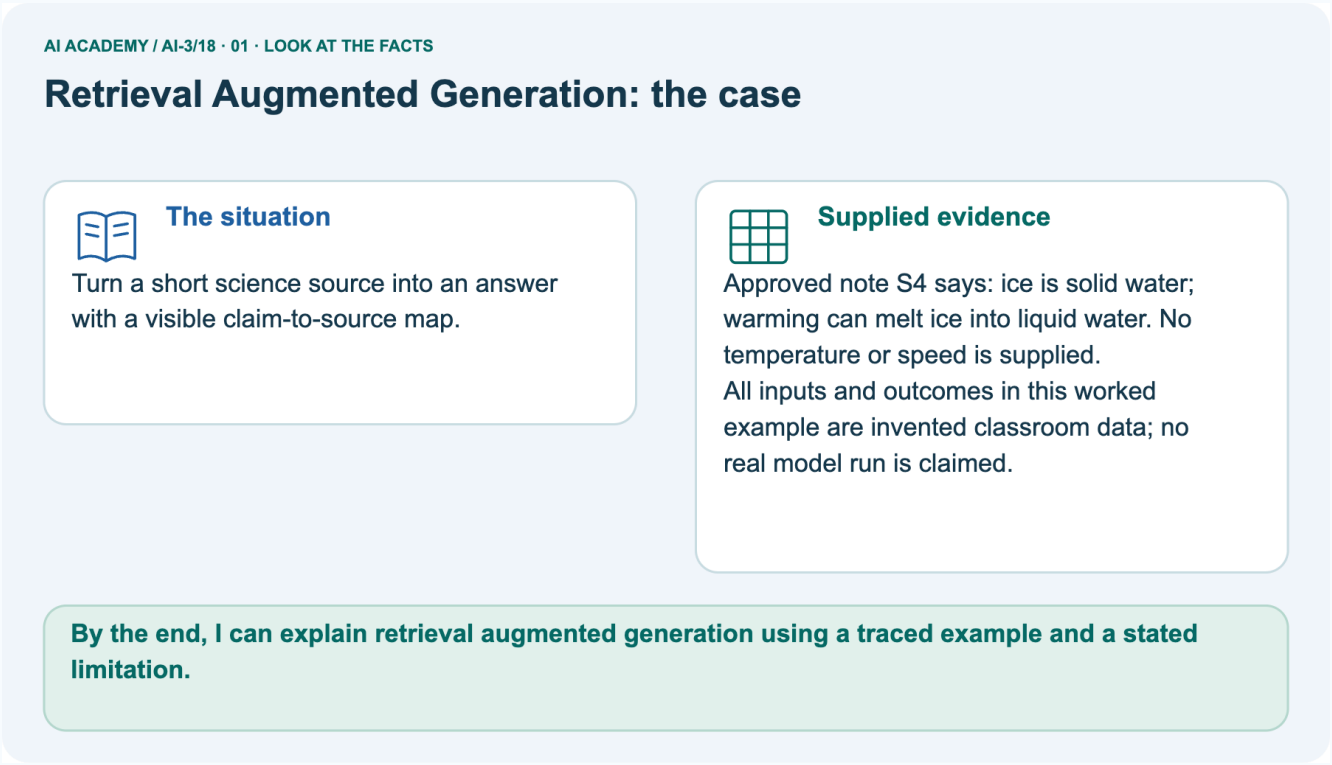
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Retrieval Augmented Generation: the case



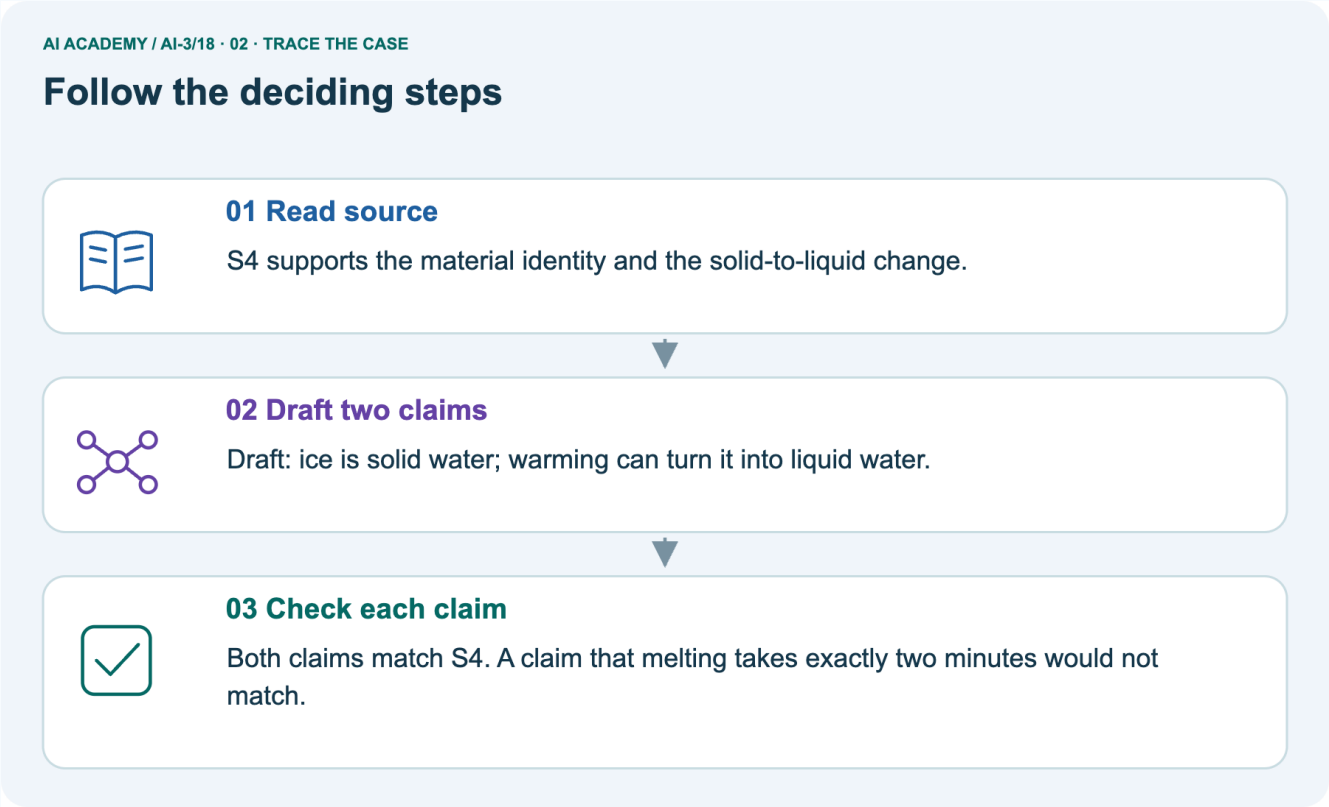
Follow the labelled picture with the matching explanation. Open the vector diagram

Turn a short science source into an answer with a visible claim-to-source map. Supplied facts: Approved note S4 says: ice is solid water; warming can melt ice into liquid water. No temperature or speed is supplied.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



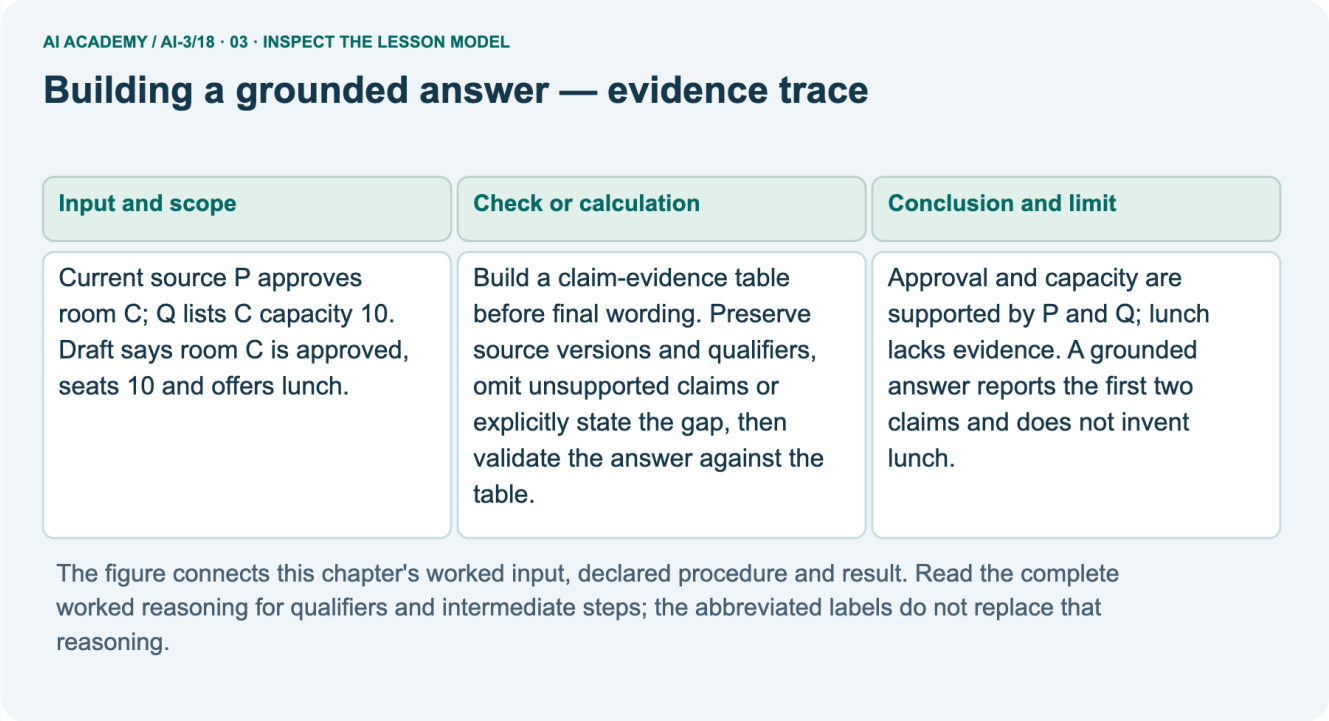
Follow the labelled picture with the matching explanation. Open the vector diagram

Read source: S4 supports the material identity and the solid-to-liquid change. → Draft two claims: Draft: ice is solid water; warming can turn it into liquid water. → Check each claim: Both claims match S4.

A claim that melting takes exactly two minutes would not match.

Point to each numbered stage and explain which recorded **input** it uses.

03. Building a grounded answer — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

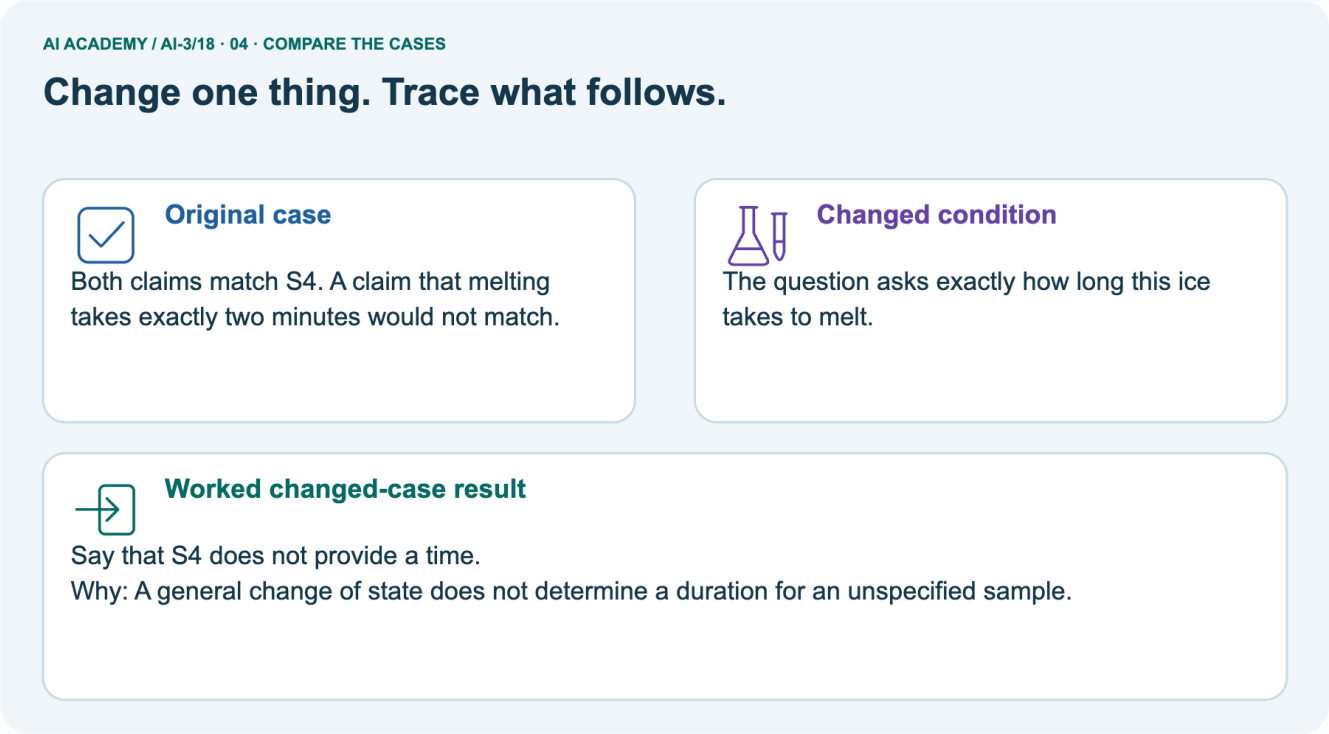
Building a **grounded answer** — **evidence** trace. Current source P approves room C; Q lists C capacity 10. Draft says room C is approved, seats 10 and offers lunch. | Build a claim-**evidence table** before final wording.

Preserve source versions and **qualifier**s, omit unsupported claims or explicitly state the gap, then validate the answer against the table. | Approval and capacity are supported by P and Q; lunch lacks **evidence**.

A **grounded answer** reports the first two claims and does not invent lunch.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



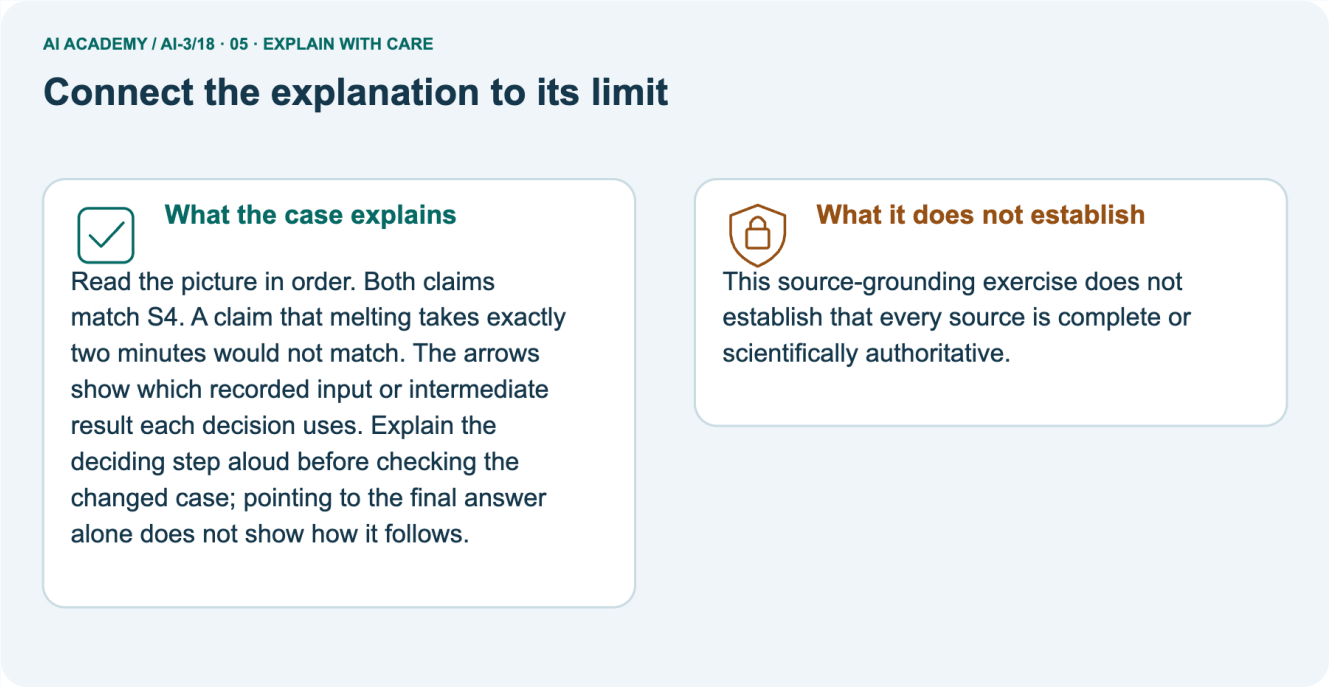
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both claims match S4. A claim that melting takes exactly two minutes would not match. Changed condition: The question asks exactly how long this ice takes to melt.

Expected result: Say that S4 does not provide a time. Reason: A general change of state does not determine a duration for an unspecified sample.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

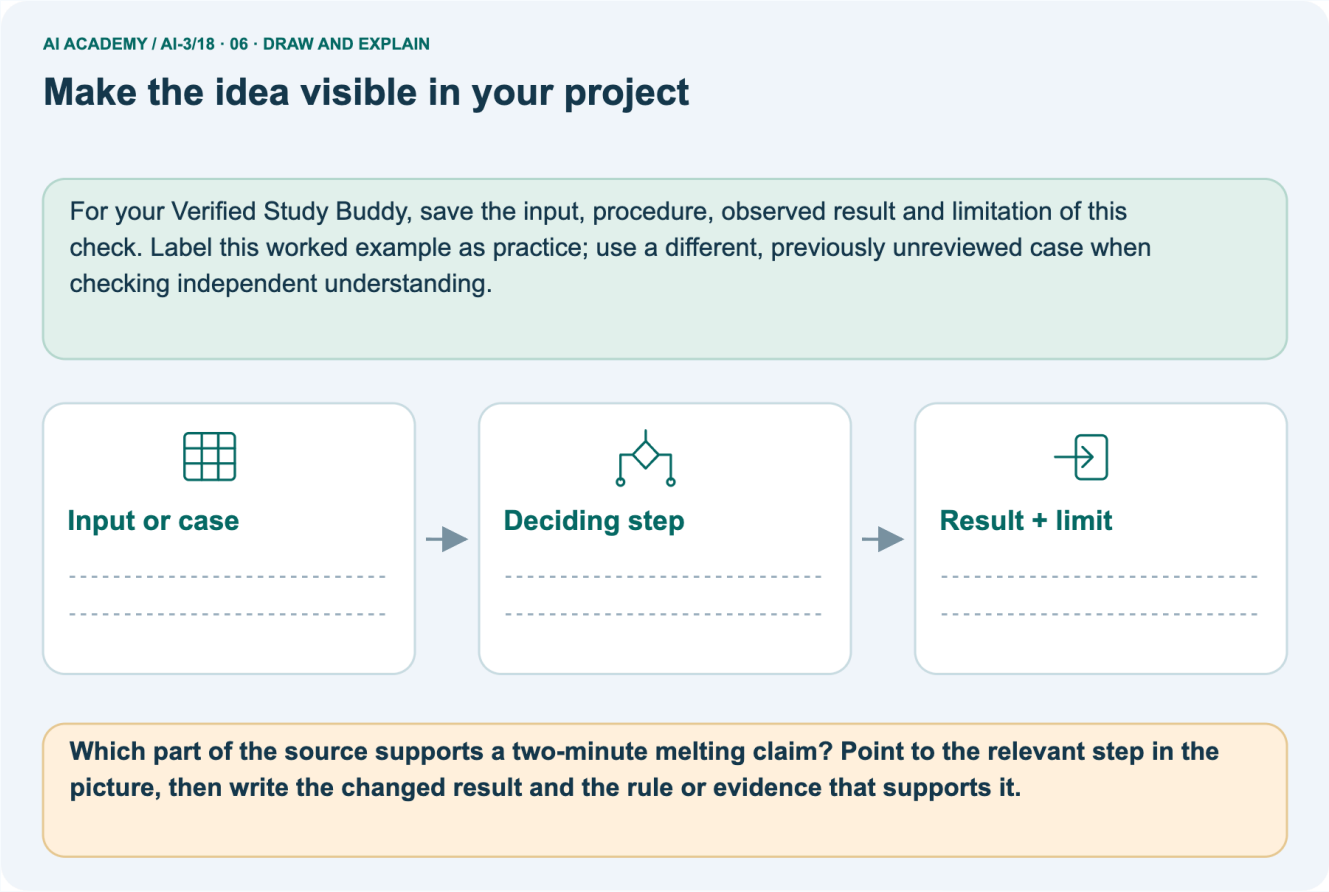
Read the picture in order. Both claims match S4. A claim that melting takes exactly two minutes would not match. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: This source-grounding exercise does not establish that every source is complete or scientifically authoritative.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part of the source supports a two-minute melting claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Retrieval Augmented Generation: the case

02. Follow the deciding steps

03. Building a grounded answer — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.